Methods of Calculation and Design for Mechanical and Public Health Building Services

How-To: Mechanical and Public Health Building Services

System Design

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Mechanical Building Services Calculation Methods

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Acronyms

ACH Air Changes per Hour.

AHU Air Handling Unit.

ASHRAE American Society of Heating, Refrigerating and Air-Conditioning Engineers.

BB101 Building Bulletin 101 - Guidelines on ventilation thermal comfort and indoor air quality in schools.

CIBSE Chartered Institution of Building Services Engineers.

CRAH Computer Room Air Handler.

CWS Cold Water Service.

GRP Glass Reinforced Plastic.

HBN Health Building Note.

HTM Health Technical Memorandum.

HWS Hot Water Service.

RH Relative Humidity.

VOC Volatile Organic Compounds.

To-Do:

Part I

Ventilation

Chapter 1

Introduction

This chapter will outline some of the basics of ventilation system design, such as Units, and Ventilation Rates.

1.1 Energy Balances

When performing design calculations on air flow rates, we need to know the energy transfer from different fluids either through coils, mixed air, or other means. The following equations describe the energy balance commonly used within Building Services design.

Equation 1.1, shows the relationship between Energy in Kilo-Watts [kW], Mass Flow Rate in kilograms per second [kg/s], Specific Heat Capacity in Joules/Gram-Kelvin [J K/g], and Temperature Difference in Kelvin [K]. Equations 1.2 through 1.4, shows the dimensional analysis of equation 1.1.

$$Q = m\ C_p\ \delta T \tag{1.1}$$

Expressing equation 1.1 as units:

$$\left[\frac{kg}{s}\right]\left[\frac{J}{g \cdot K}\right]\left[\frac{K}{1}\right] \tag{1.2}$$

Equation 1.2 can be reduced by removing Kelvin and grams from both the denominator and numerator:

$$\left[\frac{k\cancel{g}}{s}\right]\left[\frac{J}{\cancel{g} \cdot \cancel{K}}\right]\left[\frac{\cancel{K}}{1}\right] \tag{1.3}$$

This gives equation 1.4, note the definition of Watts [W] as a Joule per second [J/s]:

$$\left[\frac{1000}{s}\right]\left[\frac{J}{1}\right] = \left[\frac{1000}{1}\right]\left[\frac{W}{1}\right] = \left[\frac{kW}{1}\right] \tag{1.4}$$

See Equation 1.5 if you want to use volumetric flow rate [m^3/s, or L/s] instead of mass flow rate [kg/s].

1.2 Units

Cubic metres per second [m^3/s], litres per second [L/s], kilograms per second [kg/s], Air Changes per Hour (ACH) and so on.

- m^3/s, is used for larger volumetric air flow rates, there are a 1,000 litres in a cubic meter of air.
- L/s, is used for smaller volumetric air flow rates (Single terminal units/grilles etc), a litre is defined as one cubic decimeter.
- kg/s, is used to describe the mass flow rate of air, and is used when calculating energy balances .
- ACH, is used to describe the turnover of air in a room.

Converting from volumetric flow [m^3/s], [L/s] to mass flow [kg/s] requires the density of air [ρ] at the working temperature. Table 1.1 shows the density of air at some common temperatures.

Air Temperature °C	Air Density (ρ) kg/m^3
0	1.292
10	1.246
20	1.204
30	1.164
40	1.127
50	1.093

Table 1.1: Density of air at differing air temperatures

So by using a volumetric ventilation rate, which is commonly used to describe the outdoor air provision per person (See Section 1.4 for more information), we need to include density in Equation 1.1 so that we get sensible answers. Equation 1.1, becomes equation 1.5. (Dimensional analysis follows in Equations 1.6 through 1.8, and 1.9 through 1.11 for L/s).

$$Q = m\ \rho\ C_p\ \delta T \tag{1.5}$$

Following the reduction procedure in section 1.1, we can express equation 1.5 in units leaving equation 1.6:

$$\left[\frac{m^3}{s}\right]\left[\frac{kg}{m^3}\right]\left[\frac{J}{g \cdot K}\right]\left[\frac{K}{1}\right] \tag{1.6}$$

Now removing [m^3/s], [g], and [K] from both the denominator and numerator of equation 1.6:

$$\left[\frac{\cancel{m^3}}{s}\right]\left[\frac{k\cancel{g}}{\cancel{m^3}}\right]\left[\frac{J}{\cancel{g} \cdot \cancel{K}}\right]\left[\frac{\cancel{K}}{1}\right] \tag{1.7}$$

Leaving equation 1.8, again note the definition of Watts [W]:

$$\left[\frac{1}{s}\right]\left[\frac{1000}{1}\right]\left[\frac{J}{1}\right] = \left[\frac{1000}{1}\right]\left[\frac{W}{1}\right] = \left[\frac{kW}{1}\right] \tag{1.8}$$

The same procudre can be followed for litres per second [L/s] instead of cubic metres per second [m^3/s], noting the definition of [L/s] as a cubic decimeter, i.e. $1\ m^3 = 1,000\ l$:

$$\left[\frac{l}{s}\right]\left[\frac{kg}{m^3}\right]\left[\frac{J}{g \cdot K}\right]\left[\frac{K}{1}\right] \tag{1.9}$$

To remove both the litres from the first term numerator, and cubic meters from the second term denominator, we need to divide litres by 1,000, i.e. multiply by 1/1000.

$$\left[\frac{\cancel{l}}{s}\right]\left[\frac{1}{1000}\right]\left[\frac{k\cancel{g}}{\cancel{m^3}}\right]\left[\frac{J}{\cancel{g} \cdot \cancel{K}}\right]\left[\frac{\cancel{K}}{1}\right] \tag{1.10}$$

$$\left[\frac{1}{s}\right]\left[\frac{J}{1}\right] = \left[\frac{W}{1}\right] \tag{1.11}$$

the primary difference between using [m^3/s] and [L/s] is the addition of 1/1000 when cancelling the [m^3] and [L]; however, the 1/1000 further cancels out and the end result provides Watts instead of kilo-Watts.

1.3 Shadow Gaps and Door Undercuts

Door undercuts are used as a method of distributing air between rooms without ductwork, by simply allowing space beneath the door for air to pass then sufficient air can be introduced for a small occupied space provided that there is a method of extraction on the other side of the door. The supply and extraction provides the air flow between the rooms, and the door undercut provides the air path.

Note: *Door undercuts are not suitable for all applications, for example, if the door is a fire door, then the undercut may not be possible due to fire regulations. In this case, a transfer duct may be required to provide the necessary air flow between rooms which can be installed with a fire damper to maintain the integrity of the fire wall. See Section 5.3 for more information on fire and fire/smoke dampers.*

Shadow Gaps can provide an air path to a return air plenum, this avoids the use of return air grilles. This can provide a more aesthetically pleasing finish, but needs close coordination with the Architectural design and detailing to ensure that the gap is correctly sized and located.

The calculation of free area is the same for both door undercuts and shadow gaps. The difference being that the length of the door undercut is determined by the width of the door so only the height is adjustable, whereas the length and width of a shadow gap is adjustable. Equation 1.12, shows the calculation of free area for a given volumetric flow rate, where Q is the volumetric flow rate [m^3/s], v is the velocity of the air stream through the gap [m/s], and A is the free area [m^2].

$$Q = vA \tag{1.12}$$

When using Equation 1.12 to calculate the free area of a door undercut, then by knowing the door width, the height of the undercut can be specified. This two step calculation can be combined into one as shown in equation 1.13, where h is the undercut height [m], and w is the door width [m].

$$h = \frac{Q}{wv} \tag{1.13}$$

Sensible air velocities for shadow gaps and door undercuts are below 2.0 m/s. This avoids excess pressure, and noise generation.

1.4 Ventilation Rates

The amount of air supplied or extracted from a space differs depending on the purpose of ventilation. If air is supplied to maintain a good indoor air quality then the concentration of pollutants (Such as CO_2) in both the supply air and indoor air needs to be considered alongside the generation of pollutants, see subsection 1.4.1. Rules of thumb for the amount of outdoor air that should be provided per person are frequently given in guides such as Chartered Institution of Building Services Engineers (CIBSE) Guide A, some outdoor air rates per person and the expected quality of indoor air are given in Table 1.2.

Supplied Air Rate l/s/p	Indoor Air Quality	Classification [BS EN 13779]	Total Indoor CO_2 [ppm]
6	Low	IDA4	1600
8	Moderate	IDA3	1200
12.5	Medium	IDA2	900
20	High	IDA1	750

Table 1.2: Commonly Used Outdoor Air Rates, see CIBSE Guide A and BS EN 13779

1.4.1 Pollutant Control

Review this section

An internal space may suffer from a build-up of pollutants such as CO_2, for Volatile Organic Compounds (VOC). The levels of these pollutants within the space can be controlled using mechanical ventilation, by providing external to the space and removing some of the internal air, the concentration of the pollutant can be reduced. Equations 1.14, and 1.15, describe the build-up of a pollutant both over time, and at a steady state condition.

CO_2 is typically released through respiration, an adult human would typically release 0.005 L/s each, and outdoor air typically contains 310 ppm of CO_2 (0.000 31 L per L/s of outdoor air).

$$C_t = \frac{q_v c_0 + q_e}{q_v + q_e}\left[1 - e^{-\frac{q_v + q_e}{V}t}\right] \tag{1.14}$$

$$C_E = \frac{q_v c_0 + q_e}{q_v + q_e} \tag{1.15}$$

1.4.2 Kitchen Ventilation Rates

In the DW172 document published by the Heating and Ventilation Contractors Association (HVCA), there are five methods of determining the appropiate extract rate for kitchen extract systems. A mechanical consultant will usually have the extract rate specified by the kitchen specialist, but early stage designs specialists are not usually appointed so an estimation is usually required.

If mechanical supply air is to be provided to the space it is done so at 85% of the extract volumetric flow rate.

The thermal convection method (Method 1) is the only acceptable method for final design, the other four methods should only be used for indicative or provisional extract flow rates.

1.4.2.1 Extract Sizing - Method 1

This method uses the thermal convection method, and specifies a [m^3/s] per [m^2] of catering appliance work area. Table 2 in DW172 shows volumetric extract flow rates for different appliances, then applying a canopy factor. An example of calculating the total extract rate is given in Table 1.3.

Item	Area	Fuel	Coefficient	Flow Rate
Griddle	0.4500	Gas	0.30	0.135
Open Top Range	0.6750	Gas	0.35	0.236
Solid Top Range	0.5625	Gas	0.60	0.338
Bench	0.3750	-	0.03	0.011
Twin Fryers	0.4875	Electric	0.35	0.171
Salamander Grille	0.3000	Gas	0.75	0.225
			Sub-total Extract Volume	1.116
			Canopy Factor - *Overhead wall open at both ends*	1.25
			Volumetric Extract Rate Required	1.395 m^3/s

Table 1.3: Example method of calculating the extract flow rate using Method 1 from DW172

1.4.2.2 Extract Sizing - Method 2

If the size of the canopy is known, but not the full detail of the catering equipment being installed, then by using the face velocity method the total extract rate can be estimated.

Table 1.4 shows the velocities for different equipment types, and the fastest face velocity should be used.

Loading	Face Velocity
Light (Boiling pans etc.)	0.25 m/s
Medium (Deep fat fryers etc.)	0.35 m/s
Heavy (Chargrills etc.)	0.5 m/s

Table 1.4: Face velocity for the calculation of the extract flow rate using Method 2 from DW172

1.4.2.3 Extract Sizing - Method 3

A volumetric extract flow rate calculated by assigning a [m^3/s] value per [kW] of power input, the total air supply is then totalled.

1.4.2.4 Extract Sizing - Method 4

Method 4 is based on ACH, starting at 40 ACH for volumetric extract and increasing this if high output or densely packed equipment is used. ACH rates of up to 120 can be seen, and 40 ACH is regarded as the minimum.

1.4.2.5 Extract Sizing - Method 5

An extract flow rate is assigned per metre length of extract hood, a table in DW172 has been provided that gives extract rates for different canopy types and duty levels.

Chapter 2

Psychrometrics

Figure out how to draw psychrometric processes in LaTeX

Use my OneNote Psychrometric Notes to help here!

2.1 Psychrometric Charts

2.1.1 Psychrometric Processes

2.1.2 ADP Calculations

2.1.3 Room Condition Calculations

2.1.4 Enthalpy Calculations

2.2 Manual Psychrometric Calculations

Chapter 3

Heating and Cooling via Air

General heating and cooling requirements and calculations are covered in Section 6. The specific requirements and calculations for heating and cooling spaces via air systems are covered in this chapter.

Once the heating and cooling loads are calculated, the next step is to determine the supply air temperature and flow rate required to meet those loads.

3.1 Comfort and Efficiency

- Supply air temperatures,
- Air flow rates and velocities,
- Noise levels,
- Humidity control,
- Energy efficiency.

3.1.1 Guides and Regulations

- Health Technical Memorandum (HTM)
- Health Building Note (HBN)
- Building Regulations,
- Building Bulletin 101 - Guidelines on ventilation thermal comfort and indoor air quality in schools (BB101),

- CIBSE Guides,
- American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standards.

3.2 Calculations

3.2.1 Heating by Air

$Q = m\ C_p\ \delta T$

3.2.2 Cooling by Air

$Q = m\ \delta H$

Chapter 4

Ventilation Plant

4.1 Air Handling Unit (AHU) Components

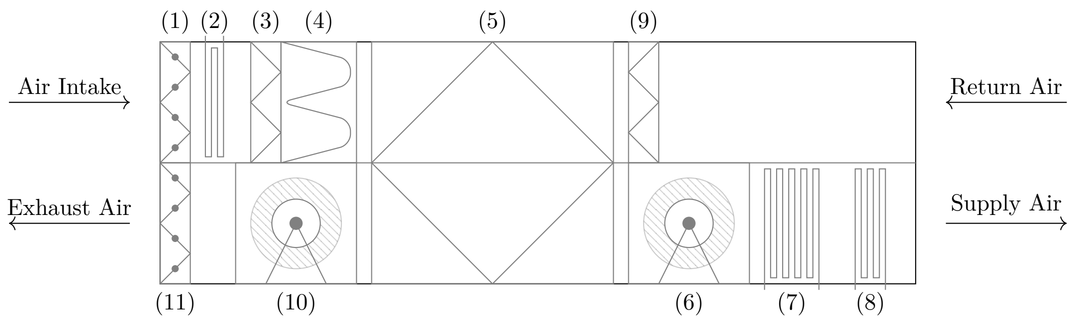

Figure 4.1: Typical AHU schematic, the component numbering follows the air path from outdoor air (Air Intake) through to the space (Supply Air), and then from the return air back to exhaust air.

A typical AHU schematic is shown in figure 4.1, (1) & (11) are motorised dampers, (2) is a frost coil, (3) & (9) are panel filters, (4) is a bag filter, (5) is a counter-flow plate heat exchanger, (6) & (10) are supply and extract fans respectively, (7) is a cooling coil, (8) is a heating or re-heat coil.

4.1.1 Energy Recovery Devices

A number of heat recovery devices are available for inclusion within an AHU.

- Plate Heat Exchanger,
- Thermal Wheel,

- Run-Around-Coil,
- Mixing Box,

4.1.1.1 Plate Heat Exchanger

The plate heat exchanger works by passing the opposing air streams (supply and extract) through fins within the unit and heat transfer occurs through the metal fins without the two air streams coming into contact with each other. See Figure 4.2 for a schematic example of a plate heat exchanger in a cross flow configuration.

4.1.1.2 Thermal Wheel

Thermal wheel units utilise a metal perforated wheel which is constantly rotating, this warms up in the extract air stream and then the warm section is rotated into the supply air stream and transfers heat to the air. Thermal wheels mix small amounts of the air streams, and involves the running of a motor to drive the wheel.

4.1.1.3 Run-Around-Coil

Run-Around-Coil units use two finned coils (similar to the heating and cooling coils) filled with a glycol/water mixture, and run the mixture between the two coils positioned in each of the air streams (supply and extract). This system utilises a pump, and associated pressurisation equipment in order to pass water/glycol between the two coils.

4.1.1.4 Mixing Box

Mixing Box units recirculate a proportion of the air streams in order to satisfy external air supply and heating/cooling loads simultaneously, for example if the air supply to support the cooling load of a space is double the outdoor air provision, then a mixing box can be used to recirculate half the cooling load air flow rate and allow the rest to come from outside. This can reduce the amount of conditioning required for the supply air as the recirculated air may be closer to the required set-point than the external air.

4.1.1.5 ErP Regulations

Under the ErP regulations[1], some form of energy recovery must be used and only plate heat exchangers, thermal wheels, or run-around-coils are acceptable. In general the regulations apply to bi-directional and uni-directional ventilation units within commercial sized buildings; however, there are exceptions for certain applications such as Swimming Pools, Agricultural Buildings, Foundries etc.

4.1.1.6 HTM 03-01 Standard AHU Layout

HTM 03-01 prescribes a standard AHU layout for healthcare applications, this layout includes a plate heat exchanger as the energy recovery device. The layout includes a fog coil, which is designed to prevent ice build-up on the filters and plate heat exchanger during the winter design condition, which is typically fully saturated air at temperatures lower than 0 °C. The layout also includes a cooling coil and a heating coil, which are used to condition the supply air to the required set point. The cooling coil is positioned before the heating coil to prevent any potential freezing within the coil; however, this precludes the ability to dehumidify the air stream should it be necessary/desired. The layout also includes a bag filter on the supply air side, and panel filters on the extract air side, this is to prevent any potential damage to the energy recovery device from dirty air, and to ensure that the supply air is clean for the space.

Include a schematic of the HTM 03-01 standard AHU layout, and reference it in the text above. Check layout against the standard, and ensure that the component numbering is consistent with the typical AHU schematic shown in Figure 4.1.

4.1.2 Energy Recovery Calculations

Equation 4.1 shows the equation describing the efficiency of the heat recovery unit (η) and the temperatures at each point for balanced air flow rates (i.e. supply air flow rate is equal to extract air flow rate), this equation is derived in Derivation 1 and shows allows for unbalanced air streams (i.e. supply air rate does not equal the extract air rate). Figure 4.2 shows a plate heat exchanger (PHX) and the appropriate labels to correspond to Equation 4.1. Minimum values prescribed in the ErP regulations for heat recovery efficiency can be found in Table 4.1.

$$\eta = \frac{t_2 - t_1}{t_3 - t_1} = \frac{t_{co} - t_{ci}}{t_{hi} - t_{ci}} \tag{4.1}$$

[1] Known by many different names: Energy related Products (ErP), Ecodesign Directive (№ 2009/125/EC), Lot 6 - Ventilation Units, Ecodesign Regulation (№ 1253/2014), UK Statutory Instrument 2015 (№ 469)

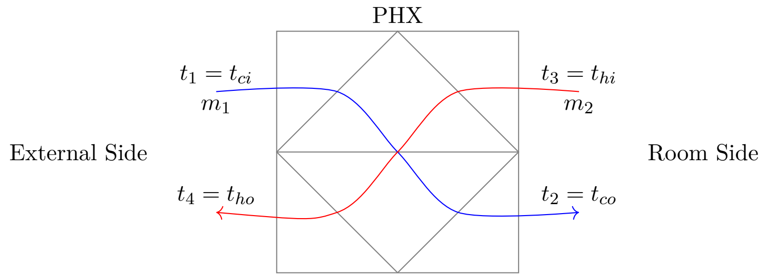

Figure 4.2: A counter-flow plate heat exchanger (PHX) showing air flow paths and temperature labels corresponding to Equation 4.1.

Heat Recovery Type	ErP 2018 Minimum Efficiency
Plate Heat Exchanger	73%
Thermal Wheel	73%
Run-Around-Coil	68%

Table 4.1: Table showing the minimum efficiency required for the three acceptable heat recovery equipment types in air handling units

Derivation 1 *This is a derivation of the energy balance within an AHU heat recovery device allowing for unbalanced air flow rates. First we start with the basic energy balance equation defined in Equation 1.1.*

$$Q = m\ C_p\ \delta t$$

Now we consider the supply air air-path (Cold), and the extract air air-path (Hot) separately.

$$Q_{cold} = m\ C_p\ (t_{co} - t_{ci})$$

$$Q_{hot} = m\ C_p\ (t_{hi} - t_{ho})$$

Using the above energy balances we can calculate the cold out temperature (Supply Air, Room side) if we know the temperature of the other three air streams, as this is rarely the case, we need to make some assumptions

If the heat exchanger were to be infinitely long, then the temperature of the hot in and cold out positions would be equal, i.e.

$$t_{hi} = t_{co}$$

Now in order to know the air temperature leaving the heat exchanger on the supply air side (t_{co}), we would only have to know what the extract air temperature is (t_{hi}); similarly, the temperature of the hot out (t_{ho}) and cold in (t_{ci}) positions would be equal, i.e.

$$t_{ho} = t_{ci}$$

An infinitely long heat exchanger is impractical (and impossible), so we can account for the reduction in heat transfer by using an efficiency term (η) as a ratio of the energy balances, i.e.

$$\frac{Q_{1,cold}}{Q_{2,hot}} = \frac{m_1\ C_p\ (t_{co} - t_{ci})}{m_2\ C_p\ (t_{hi} - t_{ho})}$$

let $\eta \times t_{ci} = t_{ho}$

$$\eta = \frac{m_1 \cancel{C_p} (t_{co} - t_{ci})}{m_2 \cancel{C_p} (t_{hi} - t_{ci})}$$

using the labels in Figure 4.2, the equation becomes;

$$\eta = \frac{m_1}{m_2} \frac{t_2 - t_1}{t_3 - t_1}$$

QED

4.1.3 Frost Coils

During the winter ventilation design condition, which is typically fully saturated air at temperatures lower than 0 °C, the build-up of ice within an AHU is possible. Ice can build-up on filters, plate-heat exchangers, or coils that are not heated (i.e. cooling coils or in-active heating coils), but typically only when the unit is running. To combat the ice build up, frost coils can be installed which are sized with sufficient capacity to raise the temperature from the external design condition (say −5 °C) to a positive temperature (typically 5 °C).

Whilst the unit is operational it is fair to assume that the heating coil and energy recovery devices will be activated and so any damage from ice build up will be limited to the filters (shown at locations (2) and (3) in Figure 4.1), as the energy recovery device will warm the incoming air using the return air. When a filter is frozen there is a risk that the filter will break up, or be pulled through the unit by the action of the fans, which will potentially damage the energy recovery device, or downstream filters. Certain manufacturers will guarantee that their filters can be used in freezing conditions without breaking, therefore negating the need for a frost coil but bringing a component of risk.

Using a frost coil within an AHU will increase the internal static pressure required to be overcome by the fans (Increasing specific fan power (SFP)), reduce the difference in temperature across the energy recovery device (reducing the 'free' energy applied to the supply air stream[2]), and increase the cost of both the unit and installation.

If a frost coil is omitted from an AHU then additional precautions are required, for example, additional control routines would be built into the control system that would shut dampers if the unit begins to freeze and turn the heating coil to full to attempt to defrost any frozen components. If a cooling coil is present within the unit, then it would need to be sited after

[2] 'free' in the sense of recovered energy that would otherwise be lost to atmosphere without the energy recovery device

the heating coil to prevent any potential freezing within the coil; however, this precludes the ability to dehumidify the air stream should it be necessary/desired.

4.1.3.1 Fog Coils

The concept of a fog coil was introduced by HTM 03-01[3] as an alternative to the standard frost coils described previously in subsection 4.1.3. The fog coil is designed to increase the incoming temperature by a fixed amount, typically, 3 °C, this moves the incoming air condition away from the dew point without the need to raise the temperature to a positive value, and therefore reduces the energy demand of the coil and the associated increase in fan power.

4.1.3.2 Calculations with and without Frost Coils

If we consider two AHU with both with a design volumetric air flow rate of $1\,\mathrm{m^3/s}$, a energy recovery device with an efficiency, η, of 73%, and one unit with a frost coil and one without, we can compare the difference in running costs. We are considering the winter time design, so external temperature is $-5\,°\mathrm{C}$, and the AHU supply temperature set point is 20°C.

Lets consider the LTHW load, Equation 4.5 shows the required LTHW mass flow rate with a frost coil at our design duty, and Equation 4.8 shows the required LTHW mass flow rate without a frost coil.

Firstly the AHU with a frost coil, we consider the energy required to heat $1\,\mathrm{m^3/s}$ of air from $-5\,°\mathrm{C}$ to $5\,°\mathrm{C}$:

$$Q = 1 \times 1.2 \times 1.005 \times (5 - -5) = 12.06kW \tag{4.2}$$

Now with a off coil temperature of 5 °C from our frost coil and a return air temperature of 20 °C we can calculate the temperature gain across our energy recovery device (Using Equation 4.1 described earlier):

$$t_2 = (0.73 \times (20 - 5)) + 5 = 15.95°\mathrm{C}, say\ 16°\mathrm{C} \tag{4.3}$$

Finally we need the energy required to heat our air stream from 16 °C to 20 °C:

$$Q = 1 \times 1.2 \times 1.005 \times (20 - 16) = 4.82kW \tag{4.4}$$

[3] HTM 03-01: Specialised ventilation for healthcare premises, Part A: Design and validation, 2018

So in total we require $12.06 + 4.82 = 16.88kW$ of LTHW input, the mass flow rate of which is:

$$m = \frac{16.88}{4.187 \times (80 - 60)} = 0.20kg/s \tag{4.5}$$

Now we consider the same AHU without the frost coil, ignoring any detriment to frozen filters etc. We know that the air stream is at −5 °C and our return air is at 20 °C, so the temperature rise across our energy recovery device is:

$$t_2 = (0.73 \times (20 - -5)) + -5 = 13.25°\text{C}, \textit{say } 13°\text{C} \tag{4.6}$$

This is only three degrees lower than our example with a frost coil, meaning that we recover a lot more heat due to the larger temperature difference, raising the temperature 18 °C compared to 6 °C with a frost coil. Now we need to calculate the required energy to heat our air stream from 13 °C to 20 °C:

$$Q = 1 \times 1.2 \times 1.005 \times (20 - 13) = 8.44kW \tag{4.7}$$

The LTHW input to this unit is:

$$m = \frac{8.44}{4.187 \times (80 - 60)} = 0.10kg/s \tag{4.8}$$

This difference in LTHW mass flow rate is vast, and when considering LTHW generation via natural gas boilers (assuming 90% efficiency) the running cost is as below:

Firstly the unit with frost coils:

$$0.20/0.9 = 0.22kg/s \tag{4.9}$$

$$Q = 0.22 \times 4.187 \times (80 - 60) = 18.42kW \tag{4.10}$$

And assuming a cost of 2.57 p/kWh, the LTHW required by the AHU with frost coils costs £11.36 per 24 hours when the external temperature is −5 °C. The AHU without frost coils:

$$0.10/0.9 = 0.11kg/s \tag{4.11}$$

Pre-Coil Type	HR Temperature (ho)	Heating Demand	Cost per 24 hours
Frost	16 °C	0.20 kg/s	£11.36
Fog	14 °C	0.13 kg/s	£7.43
None	13 °C	0.10 kg/s	£5.68

Table 4.2: Table showing differences in AHU running costs both with and without a frost coil. The AHU considered has a flow rate of $1\,\mathrm{m}^3/\mathrm{s}$, a energy recovery device with an efficiency of 73%, and a supply air set point of 20 °C. The external air temperature in this example is −5 °C and fully saturated.

$$Q = 0.11 \times 4.187 \times (80 - 60) = 9.21kW \tag{4.12}$$

Using the costs above, the LTHW generation for the AHU without frost coils costs £5.68 per 24 hours. Although the cost is relatively small this is the cost of generating LTHW to serve the AHU demand, and the difference can add up when considering the complete winter period where temperatures can be negative for a lot longer than 24 hours! This cost analysis does not consider the increase in fan running cost due to the increased resistance from the frost coil, which affects the running cost year-round whether the frost coil is active or not. Table 4.2 give a summary of the running energy and costs for an AHU with and without a frost coil, a fog coil has also been included in the table for comparison.

4.1.4 Filtration

4.1.5 Fans

4.1.6 Heating/Cooling Coils

4.1.6.1 Dehumidification

4.2 Component Resistance

Filter Resistance dirty/average/clean etc., coil resistance, PHX vs thermal wheel resistance

4.3 Fan Heat Gain

4.4 Ventilation Plant Running Costs

Chapter 5

Air Distribution

5.1 Displacement Ventilation

5.2 Ductwork Component Pressure Loss

5.3 Fire Dampers

This section will cover the application and specification of fire dampers and fire/smoke dampers. The purpose of a fire damper is to prevent the spread of fire and/or smoke through the ductwork. Fire dampers are typically installed in the ductwork at the point where it penetrates a fire-rated wall or floor. They are designed to automatically close when exposed to high temperatures, thereby preventing the passage of fire and smoke.

For the purposes of this section, fire damper references in headings will refer to both fire dampers and fire/smoke dampers, unless otherwise noted.

5.3.1 Fire Damper Types

5.3.1.1 Thermal Link Fire Dampers

5.3.1.2 Motorised Fire and Smoke Dampers

5.3.2 Fire Damper Selection

Part II

Heating and Cooling

Chapter 6

Introduction

6.1 Steady State Heating

There are several steps to calculate the required heating for a room and the size of the central plant that is required, the general procedure is as follows:

- Calculate Conduction (Fabric) Heat Loss,
- Calculate Infiltration Heat Loss,
- Calculate Ventilation System Heat Loss (If Applicable),
- Allow for Local Effects,
- Determine Intermittent Correction Factor,
- Calculate Peak Heating Load,
- Size/Select Heat Emitters,
- Calculate Distribution System Heat Loss,
- Determine Infiltration Diversity Ratio,
- Calculate Central Plant Size,
- Select Central Plant, including any redundancy.

6.1.1 Conduction (Fabric) Heat Loss

$$Q = UA \tag{6.1}$$

Define U-value here...

$$U = \frac{1}{(\sum R + R_i + R_o)} \tag{6.2}$$

Where R is the thermal resistance of the elements, R_i and R_o are the internal and external thermal resistances respectively.

Designing Buildings Wiki on U-values

6.1.2 Infiltration Heat Loss

e.g. An air pressure test result of $5\,\mathrm{m^3/(m^2\,h)}$ at $50\,\mathrm{Pa}$ would result in an air infiltration rate of approximately 0.111 ACH at atmospheric conditions.

6.1.3 Ventilation Heat Loss

6.1.4 Intermittent Correction Factor

6.1.5 Peak Heating Load

6.1.6 Distribution Losses

6.1.7 Infiltration Diversity Ratio

6.1.8 Central Plant Size

6.2 Annual Estimation of Heating Loads

6.3 Steady State Cooling

Like steady state heating, there are several steps to calculate the required cooling for a room and the size of the central plant that is required, the general procedure is as follows:

- Calculate Conduction (Fabric) Heat Gain,
- Calculate Solar Heat Gain (Transparent and Opaque),
- Calculate Infiltration Heat Gain,
- Calculate Ventilation System Heat Gain (If Applicable),

- Allow for Local Effects,
- Calculate Peak Cooling Load,
- Size/Select Cooling Units,
- Calculate Distribution System Heat Gain,
- Determine Infiltration Diversity Ratio,
- Calculate Central Plant Size,
- Select Central Plant, including any redundancy.

6.3.1 Conduction (Fabric) Heat Gain

6.3.2 Solar Heat Gain

6.3.2.1 Transparent Solar Heat Gain

6.3.2.2 Opaque Solar Heat Gain

6.3.3 Infiltration Heat Gain

Chapter 7

Evaporative Cooling

Evaporative cooling is a process that utilizes the principle of heat transfer through the evaporation of water to lower the temperature of an environment. This method is particularly effective in hot and dry climates, where the air has a low humidity level, allowing for more efficient evaporation.

Practical applications for evaporative cooling include residential and commercial cooling systems, industrial processes, and agricultural practices. In residential settings, evaporative coolers, also known as swamp coolers, are commonly used to provide a cost-effective and energy-efficient alternative to traditional air conditioning systems.

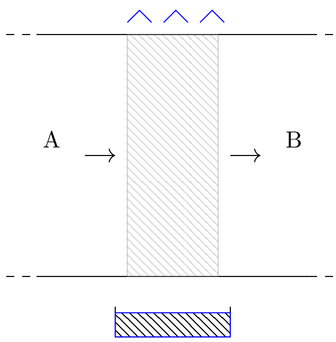

Figure 7.1: Typical schematic representation of an evaporative cooler in a duct or AHU section, the component numbering follows the air path from pre-cooler condition through to the off-cooler condition.

7.1 Principles of Evaporative Cooling

Evaporative cooling works on the principle of latent heat of vaporisation, as a psychrometric process the air is cooled along a constant wet bulb line (See Section 7.2).

Warm dry air is passed over a water-saturated medium, such as a pad or mesh. As the air flows through the medium, water evaporates into the air, absorbing heat from the surrounding environment and lowering the air temperature. The effectiveness of this process is influenced by factors such as ambient temperature, humidity levels, and airflow rates.

7.2 Psychrometric Process

INSERT PSYCHROMETRIC PROCESS DIAGRAM HERE

7.3 Calculations

Calculation of the cooling effect of evaporative coolers can be performed using equation 7.1, whilst rearranging for Θ_B as shown in 2:

$$\eta = \frac{\Theta_A - \Theta_B}{\Theta_A - \Theta_{WB}} \tag{7.1}$$

Where:

- Θ_A is the inlet temperature to the evaporative cooler
- Θ_B is the outlet temperature of the evaporative cooler
- Θ_{WB} is the wet bulb temperature of the air stream which remains constant
- η is the efficiency of the evaporative cooler *(Typically between 80-90%)*

This can be re-arranged to give the outlet temperature of the evaporative cooler:

$$\Theta_B = \Theta_A - \eta(\Theta_A - \Theta_{WB}) \tag{7.2}$$

Equation 7.2 has been derived in 2 and is used to calculate the outlet temperature of an evaporative cooler given the inlet temperature, wet bulb temperature, and efficiency of the cooler.

Derivation 2 *This is a derivation of the equation shown previously in Equation 7.1.*

First we acknowledge that the evaporative cooling is adiabatic and that the wet-bulb temperature remains constant through the evaporation process.

With an infinitely long evaporative cooler (and an infinitely powerful fan to draw the air through it!), the supply air temperature and wet-bulb temperature are equal. Therefore:

$$\Theta_B = \Theta_{WB}$$

Another way to describe this would be to use Wet Bulb Depression, this is defined as the difference between dry and wet bulb temperatures, i.e.

$$\Theta_A - \Theta_{WB}$$

As we know 100% efficiency is improbable in this situation, an efficiency ratio η can be added to the wet bulb depression equation to give the air temperature after the evaporative cooler:

$$\textit{Temperature Reduction} = \eta(\Theta_A - \Theta_{WB})$$

The achieved temperature after the evaporative cooler Θ_B would then be:

$$\Theta_B = \Theta_A - \textit{Temperature Reduction}$$

Substituting the above Temperature Reduction formula:

$$\Theta_B = \Theta_A - \eta(\Theta_A - \Theta_{WB})$$

QED

7.3.0.1 Evaporative Cooler Example

Cardiff summer design conditions (0.4%): 26.2 °C dry-bulb and 18.2 °C wet-bulb.

Wet Bulb Depression at design conditions would be:

$\Theta_A - \Theta_{WB} = 26.2 - 18.2 = 8$

i.e. A 100% efficient evaporative cooler would produce a 8 °C temperature drop at Cardiff design conditions.

Using the formula derived in 2 and shown in Equation 7.2 can provide the supply air temperature possible with a commercially available evaporative cooler assuming an evaporative cooler efficiency of 85%:

$\Theta_B = 26.2 - 0.85(26.2 - 18.2) = 19.4\,°\mathrm{C}$

7.4 Commercial Applications

The following subsections provide an overview of the different types of evaporative cooling systems used in commercial applications, including stand-alone evaporative coolers and AHU (Air Handling Unit) evaporative cooling systems.

7.4.1 Stand-alone Evaporative Coolers

Sometimes nicknamed *'swamp coolers'* or known as direct evaporative cooling, stand-alone evaporative coolers are self-contained units that can be placed in a room or space to provide cooling. These units typically consist of a fan, water reservoir, and a water-saturated medium through which air is drawn. As the air passes through the medium, it is cooled by the evaporation of water. These can be effective in dry climates, but in a sealed room will quickly increase the relative humidity to uncomfortable levels. Their effectiveness is limited in humid climates, where the air already contains a significant amount of moisture; however, in the correct conditions they can provide a cost-effective and energy-efficient cooling solution.

7.4.2 AHU Evaporative Cooling

Evaporative units can be installed within an AHU to cool the supply air to the space, cooling the supply air is the ultimate aim, but the evaporative cooling process can be implemented in different locations with various benefits and drawbacks. The following subsections provide an overview of the different types of AHU evaporative cooling systems.

7.4.2.1 Direct AHU Evaporative Cooling

The evaporative cooler is installed directly in the supply air stream of the AHU, cooling the air before it is distributed to the space. This method is effective in reducing the supply air temperature, but it can also increase the humidity of the air, which may not be desirable in certain applications. Direct AHU evaporative cooling is most effective in dry climates where the increase in humidity can be beneficial.

WORKED EXAMPLE OF DIRECT AHU EVAP COOLING HERE

7.4.2.2 Indirect AHU Evaporative Cooling

This is a cooling only system where outside air is not provided to the space by the AHU. Outside air is passed through an evaporative cooler then over a heat exchanger, and is exhausted. Indoor return air is passed over the heat exchanger, is cooled, then returned to the space.

This is not likely to be a practical method of comfort cooling within the United Kingdom given the relatively humid weather and the need to provide outdoor air to spaces to allow occupants to occupy them safely. There will be specialist use cases for this type of cooler such as data centres, storage facilities, etc where high volumes of outdoor air are not required; however, these types of facilities typically utilise closed loop refrigerant based cooling systems with specialised indoor cooling units such as COMPUTER ROOM AIR HANDLER (CRAH) units.

7.4.2.3 Exhaust Air Stream Evaporative Cooling

The evaporative cooler is installed in the return air section of the AHU allowing the return air stream to be cooled. The air is then passed through a heat exchanger cooling the supply air without the increase in humidity levels as the evaporated water is exhausted to atmosphere once passed through the heat exchanger.

Worked Example

Outside Air conditions: 26.2 °C dry-bulb and 18.2 °C wet-bulb. Room conditions: 22 °C dry-bulb and 60% RELATIVE HUMIDITY (RH)

The room conditions indicate a wet-bulb temperature of 15.5 °C which will be used later to calculate the evaporative cooling effect.

Using Equation 7.2, the return air is passed through an 85% efficient evaporative cooler and its temperature is reduced to 16.48 °C as shown in Equation 7.3. The RH of the return air stream is increased to 95%.

$$\Theta_B = 22 - 0.85(22 - 15.5) = 16.475 \tag{7.3}$$

This cooled return air is then passed through the AHU heat exchanger, cooling the supply air to the space without increasing the humidity levels. The supply air temperature can be calculated using the heat exchanger effectiveness and the temperature difference between the cooled return air and the outside air as described in subsection 4.1.2.

The corresponding supply air temperature after the heat exchanger (Assumed 73% efficient) would be 19.1 °C, as shown in Equation 7.4. Using the psychrometric chart, the supply air to the space would be 19.1 °C dry-bulb and 15.8 °C wet-bulb, which is a reduced humidity level when compared to direct AHU evaporative cooling and is approximately 80-85% RH.

$$t_2 = (0.73 \times (16.475 - 26.2)) + 26.2 = 19.1°\mathrm{C},\ \text{say } 19\,°\mathrm{C} \tag{7.4}$$

Part III

Domestic Water Services

Chapter 8

Water Tanks

This section will cover the different types of tanks that are commonly used in domestic water systems, as well as the sizing and specification of tanks, and the design of tank inlets and outlets. The section will also cover the integration of solar hot water tanks into domestic water systems, and the use of feed and expansion tanks in domestic water systems.

8.1 Tank Types

8.1.1 Sectional Tanks

Sectional tanks are made up of panels that are bolted together on site. The panels are typically made from Glass Reinforced Plastic (GRP), and the bolts are usually stainless steel. The panels are usually coated with an epoxy resin to protect against corrosion, and the inside of the tank is usually lined with a food grade epoxy resin to prevent contamination of the water. The panels are usually manufactured in a factory and then transported to site, where they are assembled. The panels are usually bolted together using a gasket to create a watertight seal. The bolts are usually tightened to a specific torque to ensure that the gasket is compressed enough to create a watertight seal.

Sectional tanks come in externally flanged and internally flanged arrangements, the difference being that externally flanged tanks have the bolts on the outside of the tank, whereas internally flanged tanks have the bolts on the inside of the tank. Externally flanged tanks are typically easier to install and maintain, as the bolts are more accessible, however they can be more prone to corrosion as the bolts are exposed to the elements. Internally flanged tanks are typically more expensive to manufacture and install, but they can be fitted in tighter spaces as access to the outside of the tank is not required.

Sectional tanks usually require levelling steels as a base to be installed upon, these are typically a minimum of 50 mm high. Alternatively, a concrete base can be used, however this is more expensive and time consuming to install. The levelling steels or concrete base must be level and flat in order to ensure that the tank is level and that the panels are properly aligned when bolted together.

8.1.1.1 Externally Flanged

8.1.1.2 Internally Flanged

8.1.2 One piece GRP tanks

8.1.3 Underground Tanks

8.1.4 Cold Water Service (CWS) Tanks

8.1.5 Hot Water Service (HWS) Tanks

8.1.6 Sprinkler Tanks

Sprinkler tanks may need to conform to LPCB requirements. In terms of sizing and specifying a tank, this means that a raised ball-valve chamber is required in order to maintain a class AB air gap.

8.2 Tank Sizing

See Domestic Water System Calculator - Excel Sheet

8.2.1 Tanks in Small Spaces

Sometimes a tank of a certain capacity is required but the space to install a tank is limited. The following subsections detail the usually allowed dimensions and sizes, alongside alternatives that may enable a tank to squeeze into a tight space.

8.2.1.1 Access Space

Usually 600 mm is allowed as a minimum around the tank in order to tighten the bolts on an externally flanged tank; however, by using a totally internally flanged tank (described in subsubsection 8.1.1.2), the access space can be reduced to nothing and the tank installed within the corner of a room. Access is still required to the manway and valve chambers.

8.2.1.2 Raised Ball Valve Chamber

Typically a raised ball valve chamber can help with space restrictions where the room height is not the same throughout the room; for example, where the ceiling is stepped or sloped. The raised ball valve chamber can be positioned at the higher end of the room allowing the full height of the tank to be used to store water in the lower end of the room. Typical raised ball valve chambers are 450 mm high, whereas manufacturers sometimes offer smaller 'special' raised ball valve chambers such as a 250 mm high chamber from Balmoral Tanks which has side access to the valve.

8.2.1.3 Manway

Access to the internals of the tank is required for periodic cleaning when the tank is drained. Standard arrangements involve manways located on the top surface of the tank, complete with a fixed 'cat' ladder. Where height is restricted, the manway can be relocated to the side wall of the tank, with a standard ladder on both the inside and outside of the tank to allow access, one drawback is that the tank has to be drained before the manway can be opened.

8.3 Tank Inlet

Include filtering (inc. automatic backwash filter), and valves

8.4 Gravity Discharge from a tank

Discharge from a tank by gravity alone is dependant on the following; Outlet Area (A), Acceleration due to Gravity (g), Height from the outlet to the water level (H), and the Coefficient of Discharge (C_d). Figure 8.1 shows a tank open to atmosphere with a side-wall discharge, and

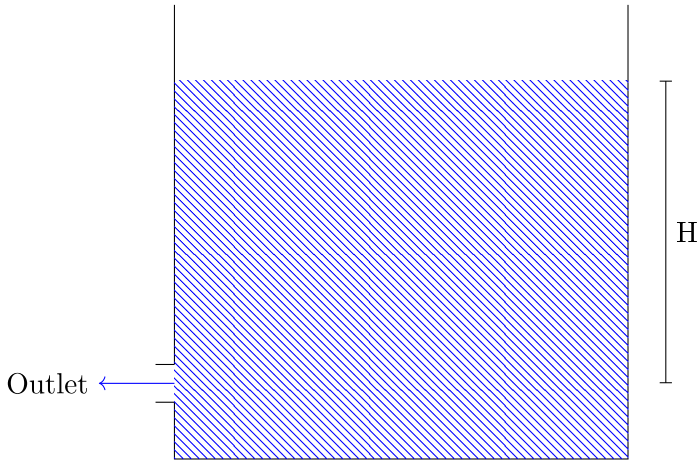

Figure 8.1: Sidewall gravity discharge from a tank

Equation 8.1 describes the volumetric flow rate through the outlet based on gravity flow, and an example tank discharge has been calculated in Equation 8.2.

$$Q = C_d A(2gH)^{\frac{1}{2}} \tag{8.1}$$

$$Q = 0.5 \times 0.00008 \times (2 \times 9.81 \times 0.8)^{0.5} = 0.00016 \text{ m}^3/\text{s} = 0.16 \text{ litre/s} \tag{8.2}$$

To calculate the centre line velocity of the discharge (U), Equation 8.3 is used, and example for the same tank has been shown in Equation 8.4. C_v has been assumed to be 0.97 in this example.

$$U = C_v(2gH)^{\frac{1}{2}} \tag{8.3}$$

$$U = 0.97 \times (2 \times 9.81 \times 0.8)^{0.5} = 3.84 \text{ m/s} \tag{8.4}$$

Finally the reaction force (F) from the water nozzle can be calculated from Equation 8.5, the density of water (ρ) has been taken as $1.000\,\text{kg/m}^3$. The example tank has been shown in Equation 8.6

$$F = \rho Q U \tag{8.5}$$

$$F = 1,000 \times 0.00016 \times 3.84 = 0.61 \text{ kg m/s}^2 = 0.61 \text{ N} \tag{8.6}$$

8.5 Solar Hot Water Tank Arrangment

See notes in OneNote link

OneNote Solar HW Integration Notes

8.6 Feed and Expansion Tank

Although no longer typically used, feed and expansion tanks were used in heating and cooling systems to provide a feed of water to the system, and to allow for expansion of the water in the system due to temperature changes. The feed and expansion tank was usually located at the highest point in the system, and was open to atmosphere. The feed and expansion tank was usually connected to the system via a ball valve, which could be used to isolate the tank from the system if necessary. The feed and expansion tank was usually sized to hold enough water to allow the system to heat up and the system water to expand from cold plus a margin to ensure the tank does not run dry. The feed and expansion tank was usually made from a material that was resistant to corrosion, such as GRP or stainless steel.

Chapter 9

Domestic Hot Water Loads

9.1 Peak Loads

9.2 Estimating Annual Hot Water Loads

Part IV

Drainage and Rainwater

Bibliography